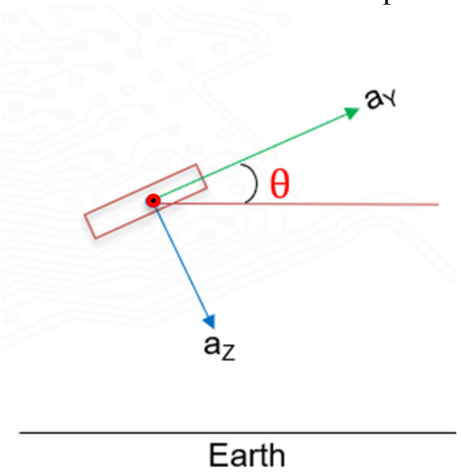
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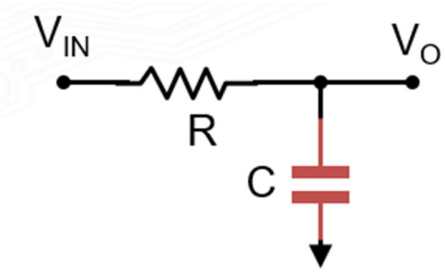
SC2104/CE3002 Sensors & Interfacing
Tutorial 2 – IMU Sensor

1. The following diagram shows the an accelerometer that is tilted along its X-axis at an angle Θ with respect to the earth frame (i.e. diagram shows the front view along its X-axis ●)

   Calculate the angle Θ in terms of the accelerometer output's a_X, a_Y and a_Z.



2. (a) Derive the expression of a digital low pass algorithm based on the basic RC low pass filter.



   (b) Find the coefficients value of the digital filter if
   (i) Bandwidth of the filter = 10 Hz
   Time interval in between samples computation (i.e. [n] and [n-1]) T = 1msec
   (ii) Bandwidth of the filter = 100 Hz
   Time interval in between samples T = 1msec

3. Describe and contrast the characteristics of Accelerometer and Gyroscope.